Source Coding

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25.08.2026

Introduction

Kraft inequality proof.

An ensemble X is a triple (x, A_X, P_X), where the outcome x is the value of a random variable, which takes one of the set of possible values $A_x = \{a_1, a_2, \ldots, a_I\}$ with corresponding probability distribution of $P_X = \{p_1, p_2, \ldots, p_I\}$ where $P(x = a_i) = p_i$

Let

$$Q = \sum_{i=1}^{I} 2^{-l_i} \tag{1}$$

The codeword for a letter/outcome indexed by i has length l_i. Consider N concatenated letters/outcomes

$$Q^N = \sum_{(i_1,\ldots,i_N)} 2^{-(l_{i_1}+l_{i_2}+\cdots+l_{i_N})} \tag{2}$$

note that the quantity in the exponent $L = l_{i_1} + l_{i_2} + \cdots + l_{i_N}$ is the codeword length of the string $\mathbf{x} = a_{i_2}a_{i_2}\ldots a_{i_N}$, for every string that can be constructed using the alphabet, there is a term in the sum. So can equivalently sum over all the possible length $Nl_{\min} \leq L \leq Nl_{\max}$, where $l_{\min}$ and $l_{\max}$ are the codeword length of the shortest codeword and longest codeword respectively, also including a factor g_L that counts how many times total codeword with length L comes up we have

$$Q^N = \sum_{L} g_L 2^{-L} \tag{3}$$

assume unique decodeability gives

$$g_L \leq 2^L \tag{4}$$

therefore

$$Q^N \leq \sum_{L=Nl_{\min}}^{Nl_{\max}} 1 \leq Nl_{\max} \tag{5}$$

the left hand side grows exponentials, the right hand side grows linearly, if $K > 1$ then there is a contradiction provides that N is sufficiently large. So we must have $K \leq 1$, and

$$\boxed{\sum_{i=1}^{I} 2^{-l_i} \leq 1} \tag{6}$$

as desired. The code is complete is the equality holds.

Given a set of codeword lengths that satisfy the Kraft inequality, there exists a uniquely decodable prefix code with these codeword lengths.

Gibbs inequality proof.
Note that

$$\ln x \leq x - 1 \tag{7}$$

let $x = q_i/p_i$ so

$$\ln \frac{q_i}{p_i} \leq \frac{q_i}{p_i} - 1 \tag{8}$$

$$p_i \ln \frac{q_i}{p_i} \leq q_i - p_i \tag{9}$$

$$\sum_i p_i \ln \frac{q_i}{p_i} \leq \sum_i (q_i - p_i) = 0 \tag{10}$$

thus

$$\boxed{\sum_i p_i \ln \frac{p_i}{q_i} \geq 0} \tag{11}$$

as desired. The relative entropy or Kullback–Leibler divergence is defined as

$$D_{\mathrm{KL}}(p||q) = \sum_i p_i \log \frac{p_i}{q_i} \tag{12}$$

and it measures how close the two probability distributions $\{p_i\}$ and $\{q_i\}$ are, and vanishes when $p_i = q_i$.

Gibbs-Bogoliubov inequality proof. Let the true probability be

$$p_i = \frac{e^{-\beta E_i}}{Q} \tag{13}$$

where $\beta \equiv k_B T$, we wish to approximate probability distribution $\{p_i\}$ by $\{q_i\}$ using a tunable set of parameters $\{\theta\}$ so $q_i \equiv q_i(\theta)$

$$\begin{aligned} 0 &\leq \sum_i q_i \ln \frac{q_i}{p_i} \\ &= \sum_i q_i \ln q_i - \sum_i q_i \ln \frac{e^{-\beta E_i}}{Q} \\ &= \sum_i q_i \ln q_i + \beta \sum_i q_i E_i + \ln Q \end{aligned}$$

then define the thermodynamic entropy as

$$S_\theta \equiv -k_B \sum_i q_i(\theta) \ln q_i(\theta) \tag{14}$$

internal energy as

$$U_\theta \equiv \sum_i q_i(\theta) E_i \tag{15}$$

and Helmholtz free energy as

$$A \equiv -k_B T \ln Q \tag{16}$$

then finally

$$A \leq U_\theta - T S_\theta \tag{17}$$

as desired.

Shannon's source coding theorem proof (For symbol codes)
A (binary) symbol code C maps the alphabet A_X to a list of 0 and 1.
The information content for an event with probability p_i is defined as

$$s_i \equiv -\log_2 p_i \tag{18}$$

and entropy is defined as the expected information content

$$\langle s \rangle \equiv -\sum_i p_i \log_2 p_i \tag{19}$$

for codeword length of l_i, define an implicit probability distribution q_i

$$q_i \equiv \frac{1}{Q} 2^{-l_i} \tag{20}$$

where the normalisation constant (like a partition function)

$$Q = \sum_i 2^{-l_i} \tag{21}$$

ensures $\sum_i q_i = 1$, thus can be interpreted as a probability distribution. So

$$\log_2 q_i = -l_i - \log_2 Q \tag{22}$$

$$l_i = -\log_2 q_i - \log_2 Q \tag{23}$$

so average length of codeword is

$$\begin{aligned}
\langle l \rangle &= \sum_i p_i l_i \\
&= -\sum_i p_i (\log_2 q_i + \log_2 Q) \\
&= -\sum_i p_i \log_2 p_i + \sum_i p_i \log_2 p_i - \sum_i p_i \log_2 q_i - \log_2 Q \\
&= \langle s \rangle + \sum_i p_i \log_2 \frac{p_i}{q_i} - \log_2 Q \equiv \langle s \rangle + \tilde{A} \\
&\geq \langle s \rangle + \sum_i p_i \log_2 \frac{p_i}{q_i} \\
&\geq \langle s \rangle
\end{aligned} \tag{24}$$

where we used Kraft inequality for the passage of line 4 to 5, and Gibbs inequality from line 5 to line 6. We have also defined the variational free energy $\tilde{A}$ on line 4.

We have shown that average codeword length cannot be lower than average expected information/entropy, ideal codeword length is the information content ($l_i^* = h_i$). The difference in $\langle l \rangle$ and $\langle h \rangle$ is due to the code being incomplete ($Q < 1$) and the difference between the true probability distribution p_i and implicit probability distribution q_i which the code with codelength l_i would be ideal for. Lastly, we wish to find the optimum code, set $l_i^* = \lceil \log_2(1/p_i) \rceil$ thus

$$\langle l^* \rangle = \sum_i p_i \lceil \log_2(1/p_i) \rceil < \sum_i p_i (\log_2(1/p_i) + 1) = \langle s \rangle + 1 \tag{25}$$

so the best we can do is to compress within one bit above entropy. We have the source coding theorem as desired

$$\boxed{\langle s \rangle \leq \langle l^* \rangle < \langle s \rangle + 1} \tag{26}$$